

urement data and images, and mostly favours the Windows operating system. But some oscilloscope makers are providing support for Linux, Mac OS X and even for Raspberry Pi.

Interestingly, apart from these, engineers can now perform analysis on their iPhones and on small devices such as a wristwatch scope (see the box titled 'Exciting innovations in the last one year.')

There are various other pocket-friendly oscilloscope devices available in the market for engineers. At the end of the day, it all depends on the engineer's preference, purpose and understanding of the devices' limitations. People will either love it or remain nonchalant with these innovations.

Knobs vs smart touch

Today, smart touch interfaces can be used to define areas where you want the oscilloscope to trigger. These interfaces are fully configurable. Traditional analogue scopes were designed for operation using dedicated knobs and buttons. However, the latest digital scopes incorporate touch-screen displays, enabling new operating concepts and enhancing ease of use, informs Appalla. He says, "As the model acceptance heavily depends on the user's taste and experience, it is advantageous if instruments support multiple ways of operation—dedicated knobs, USB keyboard/mouse and touch-screen interface. An intuitive operating interface helps users to navigate faster and reduces time in setting up, executing and documenting measurements."

Oscilloscope makers keep adding features to their instruments based on user demand for more signal analysis. "Good touch-screen implementations support drag-and-drop functionality for organising the display with multiple waveform diagrams and for selection of predefined operations. Other innovative user-interface concepts include colour-coding of buttons and knobs for guiding the user," adds Appalla. In future, more touch-screen functionality, which is popular in consumer devices, will follow.

Notable changes in oscilloscopes in last one year

- USB 3.0 in PC-based oscilloscopes
- Integration of multiple instruments in one oscilloscope
- Smart touch-screen interface
- Support for operating systems like Linux and Mac

Sharing her personal experience, Aparo notes, "Although knobs and buttons are somewhat comforting when I start working with a new oscilloscope, I have to say that moving away from traditional knobs has made it easier to automate certain tests, and it has made it possible to save common configurations. The learning curve for using a new oscilloscope with multiple menu levels is higher than when purely analogue oscilloscopes were the only option—but this is also because the new oscilloscopes can do so much more." She adds, "Obviously, having USB interfaces on oscilloscopes makes it easier to store test results and import them into other programs for writing reports or presentations."

Although oscilloscope screens are evolving to capacitive touch interfaces with multi-touch capabilities that provide zoom capability and analysis capability with great ease, Bhatia believes that common traditional knobs like channel scaling, offset and trigger modes, to name a few, are being retained as the engineers are used to operating scopes with these knobs.

As the touch has been successfully implemented in mobile systems, more electronic devices will feature this technology. Sharma says, "In this fashion, a traditional hand-held oscilloscope is being upgraded to the full touch-screen hand-held scope as a next-generation transition. This may minimise the gap between high-end and low-end scopes and bring the whole new experience of using oscilloscopes to engineers."

From 50MHz to 100GHz

An apparent factor when choosing an oscilloscope is bandwidth. And, incidentally, the biggest impact on the price of a scope is its bandwidth and the number of analogue samples per

second that it can read. For error-free results, most engineers know that the samples per second figure should be at least three to five times higher than the bandwidth. Engineers, typically, prefer

entry-level and mid-level oscilloscopes.

But almost a year back, Teledyne LeCroy had demonstrated a 100GHz real-time oscilloscope targeting applications such as CEI-25/28, CEI-56, optical coherent modulation communication systems, defense and radar applications, emerging 10-32Gb/s serial data technologies, 100GBASE-R Ethernet, SAS12, PCI Express Gen4, Thunderbolt and next-generation USB. Agilent Technologies had also introduced real-time oscilloscopes with 63GHz true analogue bandwidth in 2012. So, what's the significance of such extremely-high-bandwidth oscilloscopes?

High-speed serial standards, such as PCIe Gen4, MIPI PHY and Fibre Channel, work at speeds beyond 10Gbps, and require high bandwidth to capture the signal content, informs Bhatia. He says, "With the adoption of high-speed serial standards, bandwidth demands on oscilloscopes keep increasing." On a similar note, Aparo says, "Higher-bandwidth oscilloscopes are important in measuring the performance of higher-speed data links like USB 3.0 and JESD204B interfaces. They are also needed in the development of new high-speed digital interfaces and fibre optics."

Major significance for this kind of breakthrough is due to trends right from Telecom, RF, space technologies, to high-end medical electronics gadgets used in telemedicine, informs Mantri. However, he says, "General-purpose, lower-frequency oscilloscopes will continue to lead the market for their ease of use, lower cost and versatility. They are easy to service and find wide applications in production, service, testing and troubleshooting."

Trends and tools that help

Continuous integration at design level raises new challenges for test and